

2.4 Types of data ↕

Depending on the type of data, different types of visualisation (e.g. chart, graph, map, table, graphic) may be more effective.

Quantitative data

Quantitative data refers to data in which numerical values indicate magnitude. Addition, subtraction, multiplication and division actions can be undertaken with quantitative data.

Categorical data

Categorical data refers to data identified by labels. It is not possible to undertake mathematic operations on this data.

□ Watch: Categorical versus quantitative variables [2.36 mins]

This video explains the difference between categorical variables and quantitative variables.

This is useful because while categorical variables address how data can be grouped and numerical variables provide a measurement, it is important to recognise that numerical values can be categorical if they address how numbers can be grouped.

As you watch consider:

- How does understanding the difference between categorical variables and quantitative variables improve one's ability to analyse data efficiently and effectively?
- Why is this important in a business environment?



Source: © Nicole Hamilton (2020).

Cross-sectional data

Cross-sectional data refers to data collected at the same point in time, usually from a variety of sources.

Time series data

Time series data refers to data collected over several points in time, usually at regular intervals (minutes, hours, days, etc.).

□ Watch: Time series vs cross-sectional data [4.54 mins]

This video explains the difference between time series and cross-sectional data. This is useful for understanding that times series data can be almost any type of data that is influenced by time.

As you watch consider:

- How does the data that you interact with depend on time?
- Why is the time factor important and how does it influence the data?



Source: © Financial Analyst Tutorials (2020).

Big data

“Big data” refers to datasets that are too large and complex to be handled by standard data-processing techniques. Usually, big data is characterised by the four “V”s:

- volume – the amount of data generated
- velocity – the speed of data generation
- variety – the diversity of types and structure of the generated data
- veracity – the reliability of the generated data.

□ Watch: Big data explained [5.11 mins]

This video explains what big data is and describes a framework that can store and process it. This is useful because it enables a basic understanding of a concept that is often misunderstood.

As you watch consider:

- How could big data be useful to the organisation you work for?
- Why might big data not be advantageous for the organisation you work for?



Source: © Simplilearn (2019).

□ Activity: Why big data keeps getting bigger

Read the following article to learn about why big data keeps getting bigger.

As you read consider:

- What was useful about this article?
- What did you find difficult about this article?
- Which of the data visualisations was easier for you to understand?
- Explain your reasoning and bring your notes to the workshop.

Desjardins, J. (2019, July 16). [Why big data keeps getting bigger](https://www.visualcapitalist.com/big-data-keeps-getting-bigger/)  [\(https://www.visualcapitalist.com/big-data-keeps-getting-bigger/\)](https://www.visualcapitalist.com/big-data-keeps-getting-bigger/). *Visual Capitalist*.